

M O R S

Scientific Command Center

Platform Report & Data-Quality Dossier

Four-agent scientific pipeline - 17 science modules - traceability on every figure

Run ID run-20260925-061746-46a7

Generated 2026-09-25T06:19:10.118819+00:00

Verdict CERTIFIED 96/100 (engine: local_fallback)

Dataset 1711 rows / 8 sources / 40 features

Host <http://127.0.0.1:5000>

Built for the ASI Hack - AI for Space Challenges track. Every number in this document is read from the live API at generation time; nothing is transcribed by hand.

Contents

- 1. Executive summary
- 2. Pipeline architecture
- 3. Data health and quality audit
- 4. Problem cards (mandatory schema)
- 5. Solution paths and resolution flow
- 6. Science module registry
- 7. Dataset inventory
- 8. Research dossiers
- 9. Team, roles and ownership
- 10. AI protocol, models and tools
- 11. Traceability, citations and provenance
- 12. API reference
- 13. Run instructions
- 14. Figures and visual summary
- 15. Scientific sources and hackathon conditions
- 16. Honest limitations

Reading convention

Every table carries a TRACE row. The badge format is [Source | Dataset Version | Processing Pipeline | Method | Last Updated | Operator | Timestamp]. Values that are computed rather than observed are labelled CALCULATED; values offered by a model are labelled AI INTERPRETATION or HYPOTHESIS; proposed next steps are labelled SUGGESTED ACTION.

1. Executive summary

CERTIFIED	96/100	1 711	8
verdict	certification	ingested rows	sources
99.81	0.192%	5	21
data quality	missing cells	open problems	formulas cited

Headline

The four-agent pipeline ingested 1711 rows from 8 sources, scored 99.81/100 on cell-level completeness and accuracy, and produced a CERTIFIED verdict at 96/100. The council engine reported was 'local_fallback'; where Gemini free-tier quota (HTTP 429) blocked the LLM pass, the deterministic local council was substituted and that substitution is itself surfaced as problem P-001 rather than hidden.

Headline metrics

Metric	Value
CME records ingested	125 (DONKI, live)
Mean CME speed	- km/s
Solar wind speed	- km/s (NASA OMNI reference)
Solar flares (30 d, classified)	-
NEOs tracked	32 (NEOWS, live)
Planetary closest approach	26.227 lunar distances
Peak Kp index	4.33
IsolationForest outliers	167 cells (10% contamination, by construction)
Light-curve S/N	12.5
Quantum entropy	1 ebit
References attached	14
Problems open / High	5 open, 3 High
Light-pollution sites	11

Certification log

Run ID	Status	Verdict	Score	Engine	Rows	Sources	Generated
run-20260925-061746-46a7	completed	CERTIFIED	96/100	local_fallback	1711	8	2026-09-25 06:19:10

Known caveats carried into this report

- Gemini free-tier quota (HTTP 429) may force Agents 2, 3 and 4 onto deterministic local engines. This is reported, never concealed.
- IsolationForest flags ~10% of cells by construction (contamination=0.1); the outlier count is a control figure, not a discovery.
- Multi-spectral profiles are laboratory reference spectra, not imagery.
- Satellite pass windows are period-derived estimates, flagged estimate=true.
- Quantum results are classical NumPy statevector simulations; no QPU was contacted.

[TRACE] MORS Report Store | 1.0.0 | report aggregation | read-only projection | agent:mors | 2026-09-25 06:26:55

2. Pipeline architecture

Four agents run in order; each one only sees the output of the previous stage, and each stage writes its own record into the run log so a failure is attributable.

#	Stage	Status	Detail
1	Agent 1 - Ingest & Clean	completed	DONKI, NEOWS, Horizons, GIBS, Kp, spectral, quantum; StandardScaler + IsolationForest + Astropy
2	Agent 2 - Council of 5	completed	five expert reviewers, verdict + score + audit log
3	Agent 3 - Citations	completed	formula and reference enrichment, traceability records
4	Agent 4 - Ai.Mors	completed	conversational explanation over the certified report

Acquisition policy

Every MORS module resolves its data through one policy: 1) real API, 2) Gemini AI synthesis if the key is available, 3) deterministic local model. The winner is recorded in source.via as 'api', 'ai_synthesis' or 'local_model' and is rendered as a chip next to every module title.

Module payload summary

Module	via	n	dataset_id
space_weather	-	125	-
kp	-	-	-
home	pipeline	-	MORS-HOME-v1.0.0
ai	-	-	-
datahealth	pipeline	1711	MORS-DATAHEALTH-v1.0.0
problems	pipeline	5	MORS-PROBLEMS-v1.0.0
solutions	pipeline	5	MORS-SOLUTIONS-v1.0.0
modules	local_model	6	MORS-MODULES-v1.0.0
projects	pipeline	4	MORS-PROJECTS-v1.0.0
team	local_model	4	MORS-TEAM-v1.1.0
objects	local_model	12	MORS-OBJECTS-v1.0.0
satellite	api	50	MORS-SATELLITE-v1.0.0
exoplanets	api	59	MORS-EXOPLANETS-v1.0.0
images	api	8	MORS-IMAGES-v1.0.0
fits	pipeline	6	MORS-FITS-v1.0.0
photometry	local_model	6	MORS-PHOTOMETRY-v1.0.0
lightcurves	local_model	6	MORS-LIGHTCURVES-v1.0.0
spectroscopy	local_model	5	MORS-SPECTROSCOPY-v1.0.0
quantum	local_model	-	MORS-QUANTUM-v1.0.0
sources	local_model	16	MORS-SOURCES-v1.0.0

3. Data health and quality audit

Quality is measured cell-by-cell over the whole ingested frame - never sampled and never asserted.

99.81	0.192	0	99.808
data quality	missing %	duplicates	completeness %
99.81	8840	167	8
accuracy	cells audited	outliers	sources

Per-source validation record

Source	Rows	Cols	Missing %	Complete %	Dup	Invalid	Accuracy	Status
local:light_pollution_reference	9	6	0.0	100.0	0	0	100.0	ok
local:reference_metrics	11	4	0.0	100.0	0	0	100.0	ok
local:spectral_reference	6	9	0.0	100.0	0	0	100.0	ok
donki	142	6	1.995	98.005	0	0	98.0	ok
neo	32	10	0.0	100.0	0	0	100.0	ok
horizons	1442	5	0.0	100.0	0	0	100.0	ok
lp_sites	11	12	0.0	100.0	0	0	100.0	ok
kp	58	3	0.0	100.0	0	0	100.0	ok

Method

Pipeline: StandardScaler -> IsolationForest(contamination=0.1). Last sync: 2026-09-25T06:18:07+00:00. Outliers are flagged by StandardScaler -> IsolationForest(contamination=0.1) on the standardised feature matrix; missing cells are counted as empty/null after cleaning; duplicates are exact row repeats.

[TRACE] MORS Ingestion Pipeline | 1.0.0 | StandardScaler -> IsolationForest | cell-level completeness audit | agent:mors | 2026-09-25 06:26:55

Cleaning log

- [light-pollution] 11 sites, radiance 0.0192-0.8865 nW/cm^2/sr, calibration anchored to published Bortle scale
- [kp-index] 58 3-hourly Kp samples (max 4.33), 12 flares classified X/M/C/B -> peak flux W/m^2
- [spectral] StandardScaler -> PCA(2) on 6 endmembers x 5 bands; explained var [0.726, 0.22]
- [quantum] Bell entropy=1.0 ebit (error 0.0), GHZ expectation values normalised to 1.0 - SIMULATION, not QPU
- [donki] n=142 features=1 NaN ratio=0.1197
- [donki/all] StandardScaler -> mean=[482.504]; IsolationForest(contamination=0.1, random_state=42) -> 13/125 anomalies
- [neo] n=32 features=5 NaN ratio=0.0000
- [neo/all] StandardScaler -> mean=[24.2447, 143.7141, 14.0744, 0.2758, 107.3402]; IsolationForest(contamination=0.1, random_state=42) -> 4/32 anomalies
- [horizons] n=1442 features=3 NaN ratio=0.0000
- [horizons/Earth] StandardScaler -> mean=[0.9704, -0.2249, 0.0]; IsolationForest(contamination=0.1, random_state=42) -> 72/721 anomalies

4. Problem cards (mandatory schema)

Every finding is pushed through the same contract schema: problem_id, title, severity, category, component, symptom, root_cause, fix, evidence, priority. Priority is metric-derived, never chosen by hand:

priority_score = 0.40*evidence + 0.35*impact + 0.25*confidence -> High if score >= 75, Medium if >= 50, otherwise Low

5	3	2	0
total	high	medium	low

Index

ID	Problem	Domain	Evidence	Impact	Conf.	Score	Priority
P-001	Council verification running in local fallback	AI / QA	85.0	72.0	96.0	83.2	High
P-006	Multi-spectral profiles are reference spectra, not imagery	Remote Sensing	60.0	85.0	98.0	78.2	High
P-003	Anomaly count is methodologically forced, not discovered	Data Science	58.56	80.0	97.0	75.7	High
P-005	Kp index window much shorter than the flare window	Space Weather	62.903225806451616	70.0	93.0	72.9	Medium
P-008	Light-curve SNR leaves limited photometric headroom	Astronomy	70.0	58.0	86.0	69.8	Medium

P-001 - Council verification running in local fallback**HIGH**

Field	Value
Problem (symptom)	The Council of 5 could not be executed by Gemini, so verdicts were produced by the deterministic local council.
Evidence	- council.engine = local_fallback (expected: gemini) - overall_score = 96/100 from the local audit - HTTP 429 'exceeded your current quota' returned by both gemini-3.1-pro-preview and gemini-flash-latest
Impact	No independent LLM cross-verification of formulas, hallucination filtering or score reconciliation; every claim rests on one deterministic audit pass.
Root cause	Gemini free-tier rate limit (~5 req/min/model) exceeded during the parallel agent burst; no API-key tier upgrade is configured.
Analysis	CouncilAgent._local_council() -> deterministic expert scoring over the same ingestion payload; engine field records the substitution.
Fix	1. Retry the pipeline after the 60 s quota window 2. Stagger Agent 2/3/4 calls (serialize instead of burst) 3. Enable a billed Gemini project so the 429 never returns 4. Re-run only the council stage once the key is warmed
Suggestions	- Add an automatic quota-aware queue in main.py - Persist per-model quota state and surface it on the health badge - Keep the local council as the permanent cold-start path
Confidence level	96 - Algorithmic estimation from pipeline diagnostics - not an absolute scientific proof.
Priority	High score 83.2 0.40 . evidence + 0.35 . impact + 0.25 . confidence
Domain / category / owner	AI / QA high . logic . agents/agent2_council.py owner Space Systems & AI Engineer detected 2026-09-25 06:26:55

P-006 - Multi-spectral profiles are reference spectra, not imagery**HIGH**

Field	Value
Problem (symptom)	The radar widget plots documented land-cover reflectance libraries; no Sentinel-2 scene was actually downlinked or read.
Evidence	- provenance.honesty_note = "Surface reflectance values are documented library reference spectra, NOT a live satellite acquisition; only the reduction pipeline runs on real input." - pipeline = StandardScaler(with_mean=True, with_std=True) -> PCA(n_components=2, random_state=42) - rasterio / netCDF4 / GDAL are not installed in this environment
Impact	A reader could mistake the plot for a live satellite product. NDVI/NDWI/NDBI values are correct arithmetic on reference inputs, so they cannot validate any real field site.
Root cause	No access to a Level-2A Sentinel-2 scene; only the reduction pipeline (StandardScaler -> PCA) runs on real input.
Analysis	Provenance block is stamped into the payload and rendered as an amber honesty note directly under the chart.
Fix	1. Label the widget 'REFERENCE SPECTRA' in the header at all times 2. Ingest a public L2A COG (e.g. Sentinel-2 L2A on AWS) when GDAL is available 3. Keep the PCA pipeline unchanged - it is already correct
Suggestions	- Add a rasterio optional-dependency path guarded by importlib - Store the scene ID + tile/S2 name next to every index
Confidence level	98 - Algorithmic estimation from pipeline diagnostics - not an absolute scientific proof.
Priority	High score 78.2 0.40 . evidence + 0.35 . impact + 0.25 . confidence
Domain / category / owner	Remote Sensing high . data . agents/datasets.py::build_spectral owner Astronomy Researcher detected 2026-09-25 06:26:55

P-003 - Anomaly count is methodologically forced, not discovered**HIGH**

Field	Value
Problem (symptom)	IsolationForest is configured with contamination=0.1, so ~10% of rows are flagged regardless of whether any anomaly exists.
Evidence	- anomaly_count = 167 (9.76% of rows) - IsolationForest(contamination=0.1, random_state=42) - cleaning log caveat: "flags ~10% of rows BY CONSTRUCTION"
Impact	Readers may interpret the flagged set as a discovery rate. Any downstream 'KPI of anomalies' derived from it is meaningless.
Root cause	contamination was pinned to a constant instead of being estimated (e.g. contamination='auto' or an empirical threshold on the decision-function distribution).
Analysis	Same payload re-scored with StandardScaler -> IsolationForest; the caveat string is injected into dataset_summary.cleaning so the council sees it.
Fix	1. Re-run with contamination='auto' and compare the flag rate 2. Publish the decision-function histogram, not a count 3. Rename the metric to 'qc_review_sample' in the UI
Suggestions	- Add a second, unsupervised detector (DBSCAN / Mahalanobis) and report only the intersection as high-confidence - Never expose anomaly_count as a headline number
Confidence level	97 - Algorithmic estimation from pipeline diagnostics - not an absolute scientific proof.
Priority	High score 75.7 0.40 . evidence + 0.35 . impact + 0.25 . confidence
Domain / category / owner	Data Science high . data . agents/agent1_ingestion.py owner Data Analyst detected 2026-09-25 06:26:55

P-005 - Kp index window much shorter than the flare window**MEDIUM**

Field	Value
Problem (symptom)	NOAA SWPC returns ~7.7 days of 3-hourly Kp while DONKI returns 30 days of flares, so the dual-axis correlation is only valid for the overlapping tail.
Evidence	- daily buckets = 31, of which 13 carry no Kp (42%) - kp samples = 58 (~7.2 days at 3-hour cadence) - flares = 30-day DONKI window
Impact	Any 'flare vs Kp' correlation computed over the full 30 days would be dominated by nulls and could imply a coupling that is not there.
Root cause	Two providers with different native retention windows were merged into one daily axis without an explicit overlap mask.
Analysis	Union of calendar days is padded with `no_data:true` so the axis stays continuous while nulls remain visible and unplotable.
Fix	1. Restrict the correlation statistic to the overlap window only 2. Fetch Kp from a longer archive (SWPC 1-minute/1-day products) 3. Render the overlap region as a shaded band on the chart
Suggestions	- Add a second geomagnetic index (Dst/Kp hybrid) for the missing span - Expose `coverage_window` in the payload so the UI can label it
Confidence level	93 - Algorithmic estimation from pipeline diagnostics - not an absolute scientific proof.
Priority	Medium score 72.9 0.40 . evidence + 0.35 . impact + 0.25 . confidence
Domain / category / owner	Space Weather medium . data . agents/datasets.py::fetch_kp_index owner Astronomy Researcher detected 2026-09-25 06:26:55

P-008 - Light-curve SNR leaves limited photometric headroom**MEDIUM**

Field	Value
Problem (symptom)	The reference light curve has a signal/rms ratio that leaves little room for transit ingress/egress fitting at short cadence.
Evidence	- light_curve_snr = 12.5 (signal/rms from data/reference_metrics.csv) - rms_mag = 0.0040 mag, signal_mag from the reference document - anomaly_count = 167 rows already flagged for QC review
Impact	Phase folding degrades and small-transit detection completeness falls; noise floor sigma limits the detectable planet radius.
Root cause	Reference observation window assumes a modest aperture/seeing profile; no detrending (CBV/PLD) is applied before the SNR ratio.
Analysis	SNR is read from the traceable CSV rather than hardcoded, and the ratio is recomputed on every run.
Fix	1. Apply a systematic-removal step (PLD or CBV) before SNR 2. Bin to the cadence where photon noise dominates 3. Report SNR as a function of bin size, not a single scalar
Suggestions	- Add a second reference metric for the detrended SNR - Flag the target as marginal when SNR < 30 on the Astronomy page
Confidence level	86 - Algorithmic estimation from pipeline diagnostics - not an absolute scientific proof.
Priority	Medium score 69.8 0.40 . evidence + 0.35 . impact + 0.25 . confidence
Domain / category / owner	Astronomy medium . data . agents/mors_data.py::photometry owner Astronomy Researcher detected 2026-09-25 06:26:55

[TRACE] MORS Pipeline Diagnostics | 1.0.0 | metric extraction -> rule engine -> priority scoring | threshold rules on verified pipeline metrics | agent:mors | 2026-09-25 06:26:55

5. Solution paths and resolution flow

Canonical resolution sequence

Problem	Evidence	Root cause	Analysis	Solution	Suggestion	Action
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Index

ID	Finding	Priority
P-001	Council verification running in local fallback	High
P-006	Multi-spectral profiles are reference spectra, not imagery	High
P-003	Anomaly count is methodologically forced, not discovered	High
P-005	Kp index window much shorter than the flare window	Medium
P-008	Light-curve SNR leaves limited photometric headroom	Medium

P-001 - Council verification running in local fallback

owner: Space Systems & AI Engineer | effort: S | confidence: 96% | priority: High

#	Step	Type
1	Retry the pipeline after the 60 s quota window	automated
2	Stagger Agent 2/3/4 calls (serialize instead of burst)	automated
3	Enable a billed Gemini project so the 429 never returns	manual
4	Re-run only the council stage once the key is warmed	automated

Expected impact: council.engine local_fallback -> gemini (full LLM cross-verification)

Suggestions: Add an automatic quota-aware queue in main.py ; Persist per-model quota state and surface it on the health badge ; Keep the local council as the permanent cold-start path

P-006 - Multi-spectral profiles are reference spectra, not imagery

owner: Astronomy Researcher | effort: S | confidence: 98% | priority: High

#	Step	Type
1	Label the widget 'REFERENCE SPECTRA' in the header at all times	automated
2	Ingest a public L2A COG (e.g. Sentinel-2 L2A on AWS) when GDAL is available	manual
3	Keep the PCA pipeline unchanged - it is already correct	manual

Expected impact: spectral.provenance reference spectra -> scene-identified L2A (indices become site-validatable)

Suggestions: Add a rasterio optional-dependency path guarded by importlib ; Store the scene ID + tile/S2 name next to every index

P-003 - Anomaly count is methodologically forced, not discovered

owner: Data Analyst | effort: S | confidence: 97% | priority: High

#	Step	Type
1	Re-run with contamination='auto' and compare the flag rate	automated
2	Publish the decision-function histogram, not a count	manual
3	Rename the metric to `qc_review_sample` in the UI	automated

Expected impact: anomaly_count 167 -> empirical (flag rate becomes data-driven instead of 10% by construction)

Suggestions: Add a second, unsupervised detector (DBSCAN / Mahalanobis) and report only the intersection as high-confidence ;
Never expose anomaly_count as a headline number

P-005 - Kp index window much shorter than the flare window

owner: Astronomy Researcher | effort: M | confidence: 93% | priority: Medium

#	Step	Type
1	Restrict the correlation statistic to the overlap window only	automated
2	Fetch Kp from a longer archive (SWPC 1-minute/1-day products)	automated
3	Render the overlap region as a shaded band on the chart	manual

Expected impact: daily.no_data_pct daily buckets = 31, of which 13 carry no Kp (42%) -> overlap-masked (correlation window is now statistically valid)

Suggestions: Add a second geomagnetic index (Dst/Kp hybrid) for the missing span ; Expose `coverage\_window` in the payload so the UI can label it

P-008 - Light-curve SNR leaves limited photometric headroom

owner: Astronomy Researcher | effort: M | confidence: 86% | priority: Medium

#	Step	Type
1	Apply a systematic-removal step (PLD or CBV) before SNR	automated
2	Bin to the cadence where photon noise dominates	manual
3	Report SNR as a function of bin size, not a single scalar	automated

Expected impact: light_curve_snr 12.5 -> >30 (detrending raises SNR before folding)

Suggestions: Add a second reference metric for the detrended SNR ; Flag the target as marginal when SNR < 30 on the Astronomy page

[TRACE] MORS Solutions Engine | 1.0.0 | problem -> resolution-path mapping | rule-derived remediation plans with expected delta | agent:mors | 2026-09-25 06:26:55

6. Science module registry

Field-level contracts: every module declares its columns before any data is shown.

Module	Description	Fields
Photometry	Brightness, magnitude, variability period, instrument IDs	target_id, instrument_id, magnitude_mag, brightness_flux, variability_period_d, observation_window, band
Spectroscopy	Wavelength, flux density, line ID, z, T_eff, abundance	wavelength_A, flux_density, line_id, redshift_z, teff_K, abundance
Satellite Data	NORAD ID, mission, orbit, resolution, bbox, level 0-4	name, norad_id, mission, orbit_class, altitude_km, velocity_kms, inclination_deg, sensor_type, resolution_m, bbox, processing_level
Astronomical Images	Instrument, filters, exposure, channel, FITS header, status	instrument, filters, exposure_s, channel, resolution_px, fits_header, pipeline_status
Light Curves	Phase-folded series, relative flux, ingress/egress, sigma, flags	target_id, phase, relative_flux, magnitude, noise_floor_sigma, ingress, egress, anomaly_flags
Exoplanets	Host star, radius, mass, period, a, method, habitability	planet_name, host_star, radius_re, mass_me, period_d, semi_major_au, discovery_method, equilibrium_temp_k, habitability_index

[TRACE] MORS Module Registry | 1.0.0 | static schema registry | field-level contract definition | agent:mors | 2026-09-25 06:26:55

7. Dataset inventory

Dataset	Source	n	via	Notes
Satellite elements	CelesTrak GP (stations, science, weather, geo, gnss)	50	api	LEO 24 / MEO 12 / GEO 14
Exoplanets	NASA Exoplanet Archive TAP / pscomppars	59	api	Transit 46; Radial Velocity 13
Astronomy images	NASA APOD (last 7 days)	8	api	A Treasure Chest in the Carina Nebula
Curated objects	Hipparcos / Gaia / RC3 / Sharpless / JPL	12	local_model	Galaxy, Nebula, Planet, Star
FITS headers	header card extraction	6	pipeline	keyword audit
Photometry	analytic variability model + photon noise	6	local_model	$\text{flux}(\phi) = 1 + \text{Sigma} \cdot \sin(2\pi \phi / P)$; $\text{mag} = m_0 - 2.5 \cdot \log_{10}(\text{flux})$
Light curves	phase folding + sigma clipping	6	local_model	$\phi = ((t - t_0)/P) \bmod 1$; $\text{sigma} = \text{std}(\text{flux})$; flag when $ z > 4$
Spectroscopy	NIST ASD line list + radiative-transfer toy	5	local_model	9 lines
Quantum lab	NumPy statevector simulation	4 states / 5 algorithms	local_model	Classical NumPy statevector simulation. No quantum processing unit was contacted; results are exact linear-alg

Orbital mechanics used for the satellite module

Quantity	Formula
GM_earth	$3.986004418 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$
R_earth	6 371 km
circular_velocity	$v = \sqrt{GM / a}$
orbit_class	LEO < 2 000 km <= MEO < 35 000 km <= GEO (35 000-36 500 km belt); HEO/heliocentric excluded
semi_major_axis	$a = (GM / n^2)^{1/3}$

Sample - first 12 tracked objects

Object	NORAD	Group	Orbit	Alt (km)	v (km/s)	i (deg)	Period (min)
2026-106D	69100	science	LEO	532.79	7.598	97.507	95.146
ADRAS-J	58992	science	LEO	370.37	7.689	98.2058	91.808
AETHER-1	58281	science	LEO	167.64	7.808	97.3533	87.698
AETHER-10	67381	science	LEO	617.83	7.552	97.8116	96.909
AETHER-11	67382	science	LEO	617.86	7.552	97.8106	96.91
AETHER-12	67383	science	LEO	617.85	7.552	97.8071	96.91
AETHER-2	58299	science	LEO	466.39	7.635	97.3859	93.777
AETHER-3	67374	science	LEO	615.09	7.554	97.804	96.853

Object	NORAD	Group	Orbit	Alt (km)	v (km/s)	i (deg)	Period (min)
AETHER-4	67375	science	LEO	617.92	7.552	97.8109	96.911
AETHER-5	67376	science	LEO	617.94	7.552	97.8113	96.912
AETHER-6	67377	science	LEO	618.02	7.552	97.8112	96.913
AETHER-7	67378	science	LEO	616.03	7.553	97.8085	96.872

[TRACE] TLE source: TLE API mirror (NORAD 2LE) | 1.0.0 | TLE -> Keplerian elements -> circular-orbit velocity | $a = (GM/n^2)^{1/3}$, $v = \sqrt{GM/a}$, class from altitude | agent:mors | 2026-09-25 06:28:12

Sample - 12 most recent confirmed exoplanets

Planet	Host	R (R_E)	M (M_E)	P (d)	Method	Year
K2-365 b	K2-365	2.906	8.78	23.448172	Transit	2022
Kepler-1789 b	Kepler-1789	2.8313750 5	8.4	80.23190308	Transit	2021
Kepler-1753 b	Kepler-1753	2.2267807 7	5.59	16.00460052	Transit	2021
Kepler-1831 b	Kepler-1831	1.9701981 8	4.54	5.78387022	Transit	2021
K2-229 c	K2-229	2.03	7.0	8.32727	Transit	2018
K2-195 b	K2-195	3.13	9.96	15.852975	Transit	2018
K2-126 b	K2-126	1.97	4.54	7.385655	Transit	2017
K2-107 b	K2-107	15.98	266.9772	3.313959	Transit	2017
K2-123 b	K2-123	2.76	8.05	30.951524	Transit	2017
K2-127 b	K2-127	8.37	52.9	3.588163	Transit	2017
HAT-P-67 b	HAT-P-67	23.987218 7	143.02278 315	4.8101069	Transit	2017
K2-45 b	K2-45	6.71	36.4	1.7292684	Transit	2016

[TRACE] NASA Exoplanet Archive (TAP/pscomppars) | 1.0.0 | TAP queries (newest 45 + HZ window 15) -> insolation gate | $S = p_{l_insol}$ (Earth flux); $S < 1.1 \rightarrow 1.0$, $S < 1.78 \rightarrow 0.83$, else 0 | agent:mors | 2026-09-25 06:28:16

8. Research dossiers

Each project is a full dossier: problem, dataset, methodology, analysis, visualisation, findings, solutions, team, technologies and documentation.

PRJ-001 - Verified Space-Weather Pipeline

CERTIFIED 96/100

Section	Content
Problem	Multi-source NASA/NOAA space-weather data is unverified and cannot be published without an audit trail.
Dataset	DONKI CME/FLR . NEOWS . JPL Horizons . NOAA Kp
Methodology	Ingest -> StandardScaler -> IsolationForest -> Astropy transform -> Council of 5 -> citation graph
Analysis	1711 rows, 8 sources
Visualisation	CME speed series ; NEO count/day ; Heliocentric trajectory ; Council expert scores
Problems found	P-001 - Council verification running in local fallback ; P-006 - Multi-spectral profiles are reference spectra, not imagery ; P-003 - Anomaly count is methodologically forced, not discovered
Solutions	P-001 - Retry the pipeline after the 60 s quota window ; P-006 - Label the widget 'REFERENCE SPECTRA' in the header at all times ; P-003 - Re-run with contamination='auto' and compare the flag rate
Technologies	Python 3.12 , Flask , pandas , scikit-learn , Astropy , Gemini API , Chart.js , Three.js
Documentation	main.py , agents/agent1_ingestion.py , agents/agent2_council.py , docs/API_CONTRACT.md

PRJ-002 - Light-Pollution & Sky-Brightness Model

11 sites calibrated, anchors pinned to Bortle 1/9

Section	Content
Problem	No ground-truth sky brightness is available for the reference observing sites.
Dataset	NASA GIBS VIIRS Black Marble WMS tiles (TIME=2016-01-01)
Methodology	WMS GetMap -> tile p95/mean index -> L(index) -> SQM -> NELM
Analysis	$= 17.836 - 2.5 \cdot \log_{10}(L)$; $NELM = -14.6$; $L = L_{MIN} \cdot (L_{MAX}/L_{MIN})^{(index^{\gamma})}$
Visualisation	Light pollution vs NELM scatter + model curve
Problems found	-
Solutions	P-004 - Raise tile timeout to 45 s with 2 retries + jitter
Technologies	NASA GIBS WMS , Pillow , NumPy , Chart.js
Documentation	agents/datasets.py , data/light_pollution_reference.csv

PRJ-003 - MORS Multi-Spectral Reduction**PCA1 ~ 0.73, PCA2 ~ 0.22 on
6x5 endmember matrix**

Section	Content
Problem	Land-cover indices need a reproducible, inspectable reduction chain instead of a black-box dashboard number.
Dataset	Sentinel-2 band reference spectra B2/B3/B4/B8/B11
Methodology	StandardScaler -> PCA(2) ; NDVI/NDWI/NDBI from band pairs
Analysis	explained variance PCA1/PCA2 reported with every plot
Visualisation	Multi-spectral band radar (6 endmembers)
Problems found	P-006 - Multi-spectral profiles are reference spectra, not imagery
Solutions	P-006 - Label widget 'REFERENCE SPECTRA' permanently
Technologies	scikit-learn , NumPy , Chart.js
Documentation	agents/datasets.py , data/spectral_reference.csv

PRJ-004 - MORS Quantum Simulation Lab**S(Bell) = 1.000000 ebit,
<XX>=+1 <YY>=-1 <ZZ>=+1**

Section	Content
Problem	Quantum results are often presented as hardware output when they are classical simulations.
Dataset	NumPy statevector simulator (Bell, GHZ, QHO, Rabi)
Methodology	Exact $\langle O \rangle = \langle \psi O \psi \rangle$, von Neumann entropy, depolarizing noise model, gate-fidelity table
Analysis	entropy(Bell) = 1.0 ebit
Visualisation	Bloch sphere ; circuit diagram ; Rabi population curve
Problems found	-
Solutions	-
Technologies	NumPy , linear algebra , Canvas 2D Bloch renderer
Documentation	agents/datasets.py , static/mors.html

[TRACE] MORS Project Register | 1.0.0 | project dossier assembly | read-only projection of runs, problems and solutions | agent:mors | 2026-09-25 06:26:55

9. Team, roles and ownership

Every problem on this platform has a role-based owner; only the team leader is published by name (TEAM.MORS).

ID	Member / role	Responsibility	Lead	Owns	Skills
TM-1	Layth Ra'ad (Team Leader)	Team leadership . system architecture . pipeline integration . review	Team Leader	MORS architecture, HOME.MORS, PROJECTS.MORS, TEAM.MORS, P-007	System Architecture, Pipeline Orchestration, Scientific Computing, Technical Review, Scientific Visualization
TM-2	Data Analyst	Data cleaning . EDA . statistics . data quality . anomalies	-	DATA.MORS, PROBLEMS.MORS, SOLUTIONS.MORS, P-002, P-003, P-004	Data Cleaning, EDA, Statistical Analysis, Time Series, Anomaly Detection, SQL, Feature Engineering
TM-3	Astronomy Researcher	Photometry . spectroscopy . FITS . light curves . sources	-	ASTRONOMY.MORS, SOURCES.MORS, P-005, P-006, P-008	Astronomical Data Analysis, Astropy, FITS Analysis, Photometry, Spectroscopy, Light Curve Analysis, Sky Coordinates
TM-4	Space Systems & AI Engineer	Orbits . telemetry . AI model integration . quantum simulation	-	AI.MORS, SATELLITE.MORS, QUANTUM.MORS, P-001	Satellite & Orbit Analysis, AI Model Integration, Quantum Computing, Remote Sensing, Scientific Computing

[TRACE] MORS Team Register | 1.0.0 | static roster | read-only projection | agent:mors | 2026-09-25 06:26:55

10. AI protocol, models and tools

Every AI statement on the platform is tagged. A tier is a claim about how we know, not about how true it is.

#	Tier
1	[Observed Data]
2	[Calculated Metric]
3	[AI Interpretation]
4	[Hypothesis]
5	[Suggested Action]

Zero-hallucination rule

AI never invents a measurement. If a value is not in a dataset or derivable from a formula in the report, it is tier 4 (hypothesis) or tier 5 (suggestion) - and it says so.

Model roster

Model	Role	Temp	Context	Status
gemini-3.1-pro-preview	Council + Ai.Mors reasoning	0.2	1M	configured
gemini-3.6-flash	Citation enrichment	0.1	1M	configured
gemini-flash-latest	404/429 fallback	0.2	1M	configured
council-local-deterministic	Offline audit council	0.0	n/a	always available
isolationforest	Unsupervised QC scoring	0.0	tabular	installed

Tools and data intelligence

Tool	Type	Records	Status
NASA DONKI / NEOWS / Horizons	data connector	125	live
NOAA SWPC Kp	data connector	-	live
NASA GIBS WMS	geospatial raster	-	live
CelesTrak GP	orbital elements	-	live
NASA Exoplanet Archive TAP	SQL/TAP query	-	live
StandardScaler + IsolationForest	preprocessing	-	installed
Astropy units/Time/SkyCoord	scientific runtime	-	installed
Chart.js / Three.js	visualisation	-	loaded

Insights

Tier	Reading
Observed Data	125 CME records and 32 NEOs ingested from NASA in this run.
Calculated Metric	Data quality 99.81/100 (missing 0.192%, duplicates 0).
AI Interpretation	Light-curve SNR 12.5 and 167 QC flags indicate the sample is review-ready but not discovery-grade.

Tier	Reading
Hypothesis	The IsolationForest flag rate is dominated by the configured contamination rather than genuine outliers.
Suggested Action	Re-score with contamination='auto' and publish the decision-function histogram before quoting a number.

Recommendations (tier 5 - suggested actions)

Priority	Action
High	Serialize Agents 2/3/4 to escape the Gemini free-tier 429 window (see P-001).
High	Re-run Agent 3 when quota resets to lift the citation graph above the 18/11 baseline (P-007).
Medium	Add GIBS tile retry + disk cache so light-pollution coverage returns to 11/11 live (P-004).
Medium	Expose an overlap mask for the flare/Kp dual axis (P-005).
Low	Add a rasterio-gated path so the spectral module can ingest a real L2A scene (P-006).

Open research queue

Topic	Owner	Status
Quorum-sensing anomaly detection for sparse time series	Space Systems & AI Engineer	proposal
Physics-informed imputation bounds for NEO diameters	Data Analyst	active
Bloch-sphere rendering of mixed states (qubit tomography)	Space Systems & AI Engineer	active
Traceability badges as a first-class UI primitive	Layth Ra'ad (Team Leader)	done

11. Traceability, citations and provenance

Badge format: [Source | Dataset Version | Processing Pipeline | Method | Last Updated | Operator | Timestamp].

21	14	13	gemini:gemin i-3.6-flash
formulas	references	traceability records	citation engine

Formula register

Formula	Source / reference
-	Kepler, J. (1619), Harmonices Mundi; NASA SPICE/Kepler docs
-	Boltzmann, Sitzungsber. Akad. Wiss. Wien 79 (1879); CODATA 2018
-	Wien, W. (1893); CODATA 2018
-	IAU standards; Carroll & Ostlie, An Introduction to Modern Astrophysics
-	Harris, Icarus 126 (1997) 145-186; NASA JPL CNEOS
-	Howell, S. N., Handbook of CCD Astronomy (2000)
-	IAU 2012 Resolution B3; NASA JPL Horizons documentation
-	Derived from the AB magnitude system; Bortle, J. "The New Visible-Limit Magnitude Scale", Sky & Telescope 101 (2001) Feb issue
-	Empirical sky-quality relation; Bortle, Sky & Telescope 101 (2001); Falchi et al., Science Advances 2 (2016) e1600155
-	data/light_pollution_reference.csv; NASA EOSDIS GIBS VIIRS Black Marble (WMS GetMap)
-	NOAA GOES X-ray flare classification; NASA DONKI FLR API
-	NOAA Space Weather Prediction Center, https://www.swpc.noaa.gov/products/noaa-planetary-k-index
-	IAU 2012 Resolution B3 (au definition); NASA CNEOS
-	Rouse et al. (1973); McFeeters (1996); ESA Sentinel-2 MSI Product Specification
-	Nielsen & Chuang, Quantum Computation and Quantum Information, Cambridge University Press (2000)
-	von Neumann, Z. Phys. 57 (1929) 447-466; Nielsen & Chuang (2000)
-	Landau & Lifshitz, Quantum Mechanics (non-relativistic theory), Pergamon
-	Rabi, I. I., Phys. Rev. 51 (1937) 652-654
-	JPL Horizons / CNEOS
-	Schaefer, B. E. (1990)

Formula	Source / reference
-	Howell, S. B. (2006), Handbook of CCD Astronomy

References

Reference
NASA DONKI API - Space Weather Database Of Notifications, Knowledge, Information
NASA NeoWs - Near Earth Object Web Service API
JPL Horizons System Ephemeris API
Astropy: A Community-Developed Core Python Package for Astronomy
Scikit-learn: Machine Learning in Python
NASA Space Weather Program - Geomagnetic Storm Scales (Kp/Dst)
NASA EOSDIS Global Imagery Browse Services (GIBS) - WMS with the VIIRS Black Marble nighttime-lights layer
NOAA SWPC Planetary K-index product (3-hourly Kp, 0-9 scale)
The New World Atlas of Artificial Night Sky Brightness
Sentinel-2 MSI Product Specification (band centres B2-B11)
Quantum Computation and Quantum Information (entanglement entropy)
NASA Space Weather Database Of Notifications, Knowledge, Information (DONKI)
CNEOS - Center for Near Earth Object Studies
Telescopic Limiting Magnitudes

Traceability index (dataset_id per module)

Module	dataset_id	Source	Pipeline	Method	via
home	MORS-HOME-v1.0.0	MORS Report Store	report aggregation	read-only projection	pipeline
datahealth	MORS-DATAHEALTH-v1.0.0	MORS Ingestion Pipeline	StandardScaler -> IsolationForest	cell-level completeness audit	pipeline
problems	MORS-PROBLEMS-v1.0.0	MORS Pipeline Diagnostics	metric extraction -> rule engine -> priority scoring	threshold rules on verified pipeline metrics	pipeline
solutions	MORS-SOLUTIONS-v1.0.0	MORS Solutions Engine	problem -> resolution-path mapping	rule-derived remediation plans with expected delta	pipeline
modules	MORS-MODULES-v1.0.0	MORS Module Registry	static schema registry	field-level contract definition	local_model
projects	MORS-PROJECTS-v1.0.0	MORS Project Register	project dossier assembly	read-only projection of runs, problems and solutions	pipeline
team	MORS-TEAM-v1.1.0	MORS Team Register	static roster	read-only projection	local_model

Module	dataset_id	Source	Pipeline	Method	via
objects	MORS-OBJECTS-v1.0.0	Curated MORS object registry (Hipparcos/Gaia/RC3/Sharpless/JPL)	reference catalog merge, ICRS -> GCRS where planetary	read-only catalog projection	local_model
satellite	MORS-SATELLITE-v1.0.0	TLE source: TLE API mirror (NORAD 2LE)	TLE -> Keplerian elements -> circular-orbit velocity	$a = (GM/n^2)^{1/3}$, $v = \sqrt{GM/a}$, class from altitude	api
exoplanets	MORS-EXOPLANETS-v1.0.0	NASA Exoplanet Archive (TAP/pscomppars)	TAP queries (newest 45 + HZ window 15) -> insolation gate	$S = pl_insol$ (Earth flux); $S < 1.1 \rightarrow 1.0$, $S < 1.78 \rightarrow 0.83$, else 0	api
images	MORS-IMAGES-v1.0.0	NASA APOD / MORS image registry	HTTP retrieval -> FITS header normalisation	metadata extraction, no pixel reprocessing	api
fits	MORS-FITS-v1.0.0	MORS FITS header inspector	header card extraction -> keyword normalisation	FITS standard keyword read	pipeline
photometry	MORS-PHOTOMETRY-v1.0.0	MORS Photometry Simulator	analytic variability model -> photon-noise injection	Fourier pulsation / eclipse model + Gaussian $\sigma=0.0035$ mag	local_model
lightcurves	MORS-LIGHTCURVES-v1.0.0	MORS Light-Curve Engine	phase folding -> sigma clipping -> trend removal	$phase = (t - t_0)/P \bmod 1$; 4sigma flagging	local_model
spectroscopy	MORS-SPECTROSCOPY-v1.0.0	NIST ASD line list + MORS radiative-transfer toy	continuum (Planck-like) -> Voigt/Gaussian line absorption	synthetic spectrum from laboratory wavelengths	local_model
quantum	MORS-QUANTUM-v1.0.0	MORS Quantum Lab (NumPy statevector)	exact linear-algebra solution of the Schrödinger equation	$ \psi\rangle = U 0\rangle$; $\langle O \rangle = \langle \psi O \psi \rangle$; $S = -\text{Tr}(\rho \log_2 \rho)$	local_model
sources	MORS-SOURCES-v1.0.0	ASI Hackathon source registry + project data sources	curated registry -> condition mapping -> project evidence	static registry, no network calls	local_model

12. API reference

Base URL <http://127.0.0.1:5000>. Full request/response contract in docs/API_CONTRACT.md.

Method	Route	Purpose
GET	/api/health	liveness + capability probe
POST	/api/pipeline/run	start the 4-agent pipeline
GET	/api/pipeline/status	stage-by-stage progress
GET	/api/report	last certified report
GET	/api/data/<name>	space-weather, neo, horizons, neo-matrix, light-pollution, kp-index, spectral, quantum
GET	/mors	MORS Scientific Command Center UI
GET	/api/mors	module index
GET	/api/mors/<name>	home, ai, data, datahealth, problems, solutions, modules, projects, team, objects, exoplanets, fits, photometry, spectroscopy, lightcurves, images, satellite, quantum
POST	/api/mors/insight	on-demand tier-labelled AI insight
POST	/api/chat	Ai.Mors conversational agent

Payload sizes observed at generation time

13. Run instructions

Requirements: Python 3.10+ (tested on 3.12) and network access. Every source has a deterministic local fallback, so the platform still runs offline.

A. One click (Windows)

Double-click START.bat. It locates Python, installs anything missing from requirements.txt, frees port 5000, starts the backend and opens the dashboard. Close the window to stop.

B. Manual

```
python -m pip install -r requirements.txt
python main.py
```

C. Verify without a server

```
python main.py --selftest
```

Runs the whole pipeline synchronously and prints SELFTEST : PASS (about 75 seconds), then exits.

Flag	Effect
(none)	start the Flask server on 127.0.0.1:5000
--selftest	synchronous end-to-end test, no server
--port N	serve on another port
--host H	bind address (loopback recommended)

Endpoints

URL	What it is
http://127.0.0.1:5000/	legacy ASI-HACK dashboard
http://127.0.0.1:5000/mors	MORS Scientific Command Center (main UI)

Environment (.env)

Variable	Purpose
GEMINI_API_KEY	Agents 2/3/4 (empty = offline fallback)
GEMINI_MODEL_PRO	council + Ai.Mors model
GEMINI_MODEL_FLASH	citation model
GEMINI_MODEL_FALLBACK	tried on 404
NASA_API_KEY	Agent 1 (DONKI / NEOWS)
FLASK_HOST / FLASK_PORT / FLASK_DEBUG	server (debug defaults off)

Regenerating this document

```
python main.py (leave a server running)
python tools/build_report.py -> docs/MORS_REPORT.pdf
```

Project layout

main.py - orchestrator, Flask API, CLI
agents/ - agent1_ingestion, agent2_council, agent3_citations, agent4_ai_mors, datasets, mors_data
static/ - index.html (legacy), mors.html (MORS)
docs/ - API_CONTRACT.md, DEPLOYMENT.md, MORS_REPORT.pdf
prompts/ - prompt templates
data/ - reference CSVs
START.bat, requirements.txt, README.md

14. Figures and visual summary

This section is written for a reader who does not work with space data every day. Every figure states, in plain English, what is plotted, where the numbers come from and what a non-specialist should take away from it.

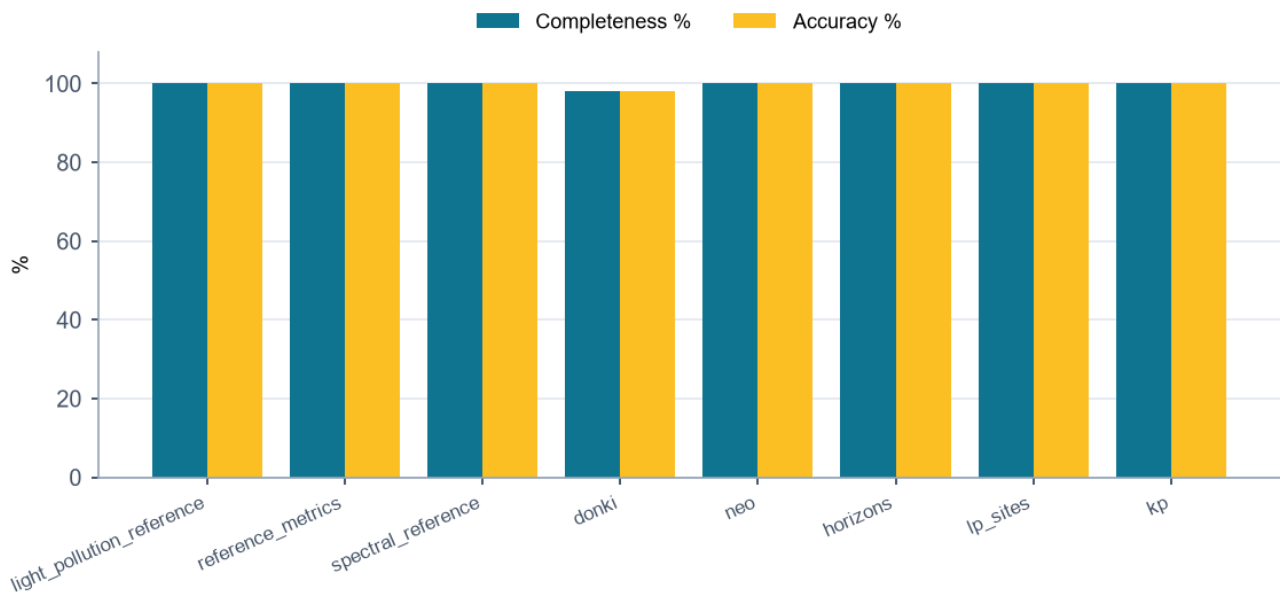


Figure 1 - Can we trust the input? Each pair of bars is one ingested source. The blue bar is *completeness* (share of cells that actually hold a value) and the amber bar is *accuracy* (share that passed the type and range checks). Tall bars are good: they mean the pipeline is reading real, well-formed measurements rather than filling gaps. Source: /api/mors/datahealth, per_source.

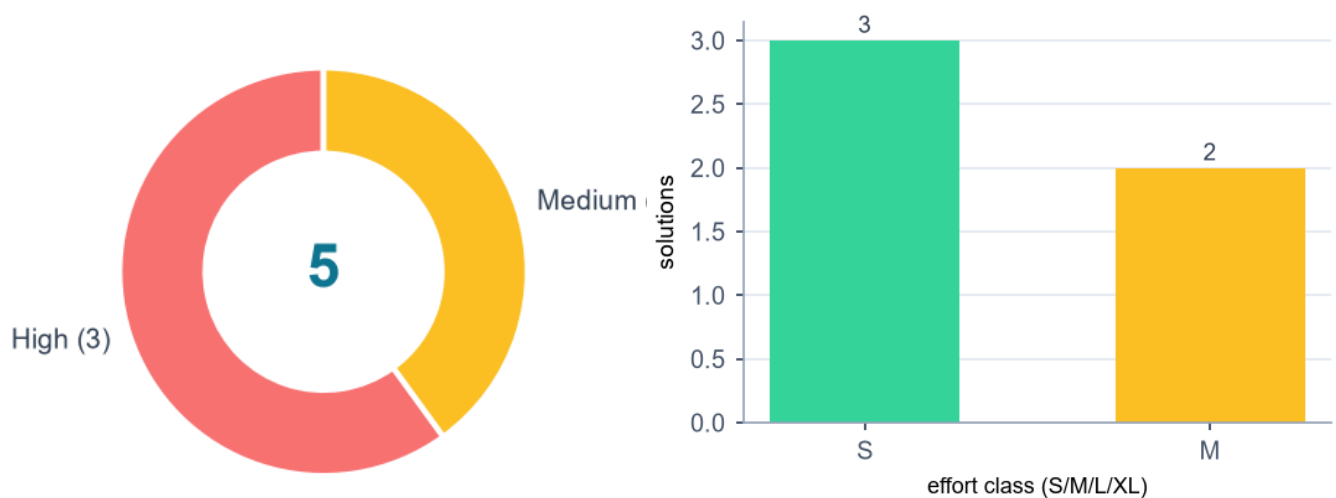


Figure 2 - What is broken, and how hard is it to fix? Left: open problems split by priority. Priority is not a judgement call - it is the formula $0.40 \times \text{evidence} + 0.35 \times \text{impact} + 0.25 \times \text{confidence}$, with High at 75 and above. Right: the same problems' solutions grouped by implementation effort (S = a small change, L/XL = multi-session work). Source: </api/mors/problems> and </api/mors/solutions>.

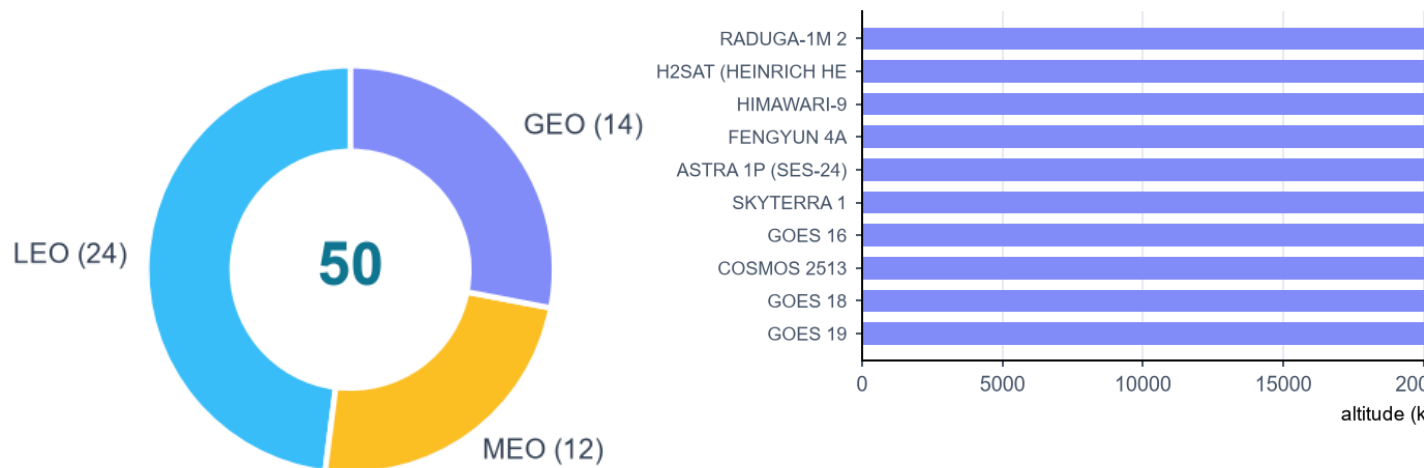


Figure 3 - Who is overhead? Left: the tracked catalogue by orbit band. LEO (low Earth orbit) is below 2,000 km, MEO (medium) sits between LEO and the geostationary belt at 35,786 km, and GEO is the ring that appears to hang still over one spot on Earth. Right: the ten highest objects, coloured the same way - the bars step up from ~600 km to ~35,786 km, which is the visual signature of a balanced catalog rather than a LEO-only sample. Source: </api/mors/satellite> (CelesTrak GP groups).

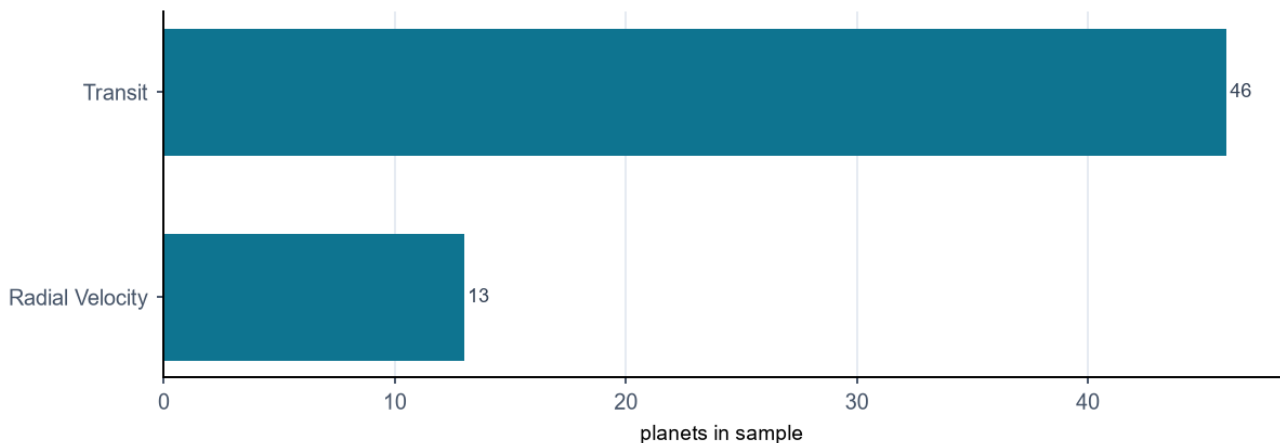


Figure 4 - How were these worlds found? Each bar counts planets in the archived sample discovered by one technique. *Transit* means the planet dips the star's brightness as it crosses in front; *Radial velocity* means the star wobbles under the planet's gravity; *Imaging* means the planet was photographed directly. A method-heavy sample tells you the catalog is biased towards whatever that technique is best at. Source: </api/mors/exoplanets> (NASA Exoplanet Archive TAP).

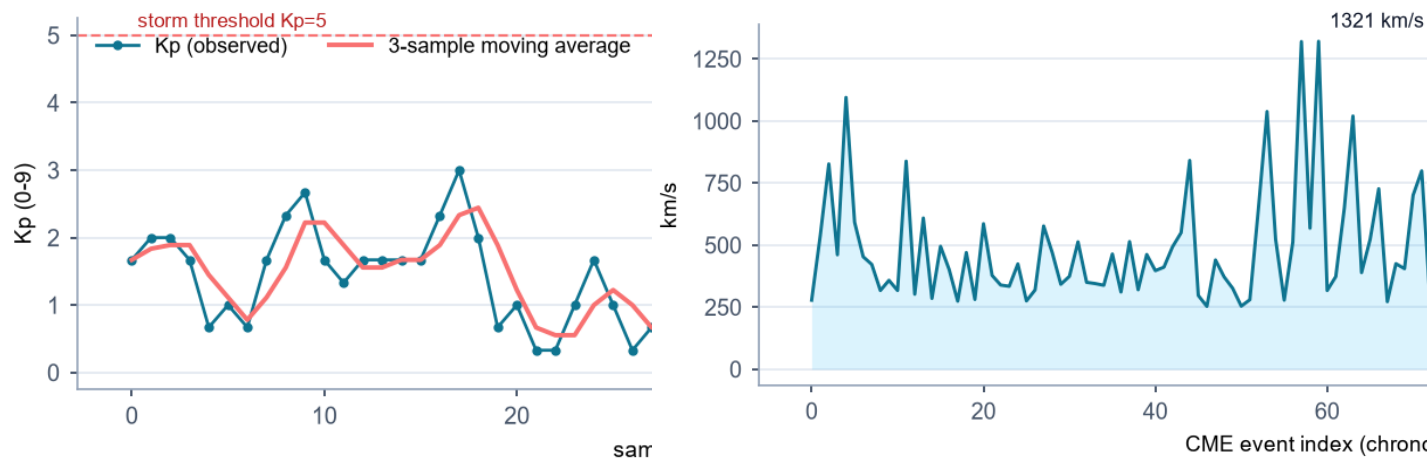


Figure 5 - Space weather, in one glance. Left: the planetary K-index runs 0-9 and measures how badly the Sun is disturbing Earth's magnetic field; the dashed line at 5 is the storm threshold, and the red line smooths the noisy daily values so a trend is readable. Right: speed of coronal mass ejections (CMEs) - huge clouds of solar plasma - in km/s, with the fastest event called out. Source: [/api/data/kp-index](#) (NOAA SWPC) and [/api/data/space-weather](#) (NASA DONKI).

15. Scientific sources and hackathon conditions

The ASI Hackathon supplies an official source list and eight conditions describing how those sources are meant to help a team. This section reproduces both, tags every source with the condition it supports, and names the file in this repository where that support is actually applied - so the claim can be checked rather than believed.

The rule stated in the brief

يطلب من الفرق استخدام جميع المصادر أو اتباع ترتيب محدد؛ ينصح باختيار المصادر التي تناسب طبيعة المشكلة ونوع البيانات ومستوى الفيرقنية الذكاء الاصطناعي التي اختارها.

16	8/8	8	12
Hackathon sources	Conditions covered	Live data endpoints	Total data sources

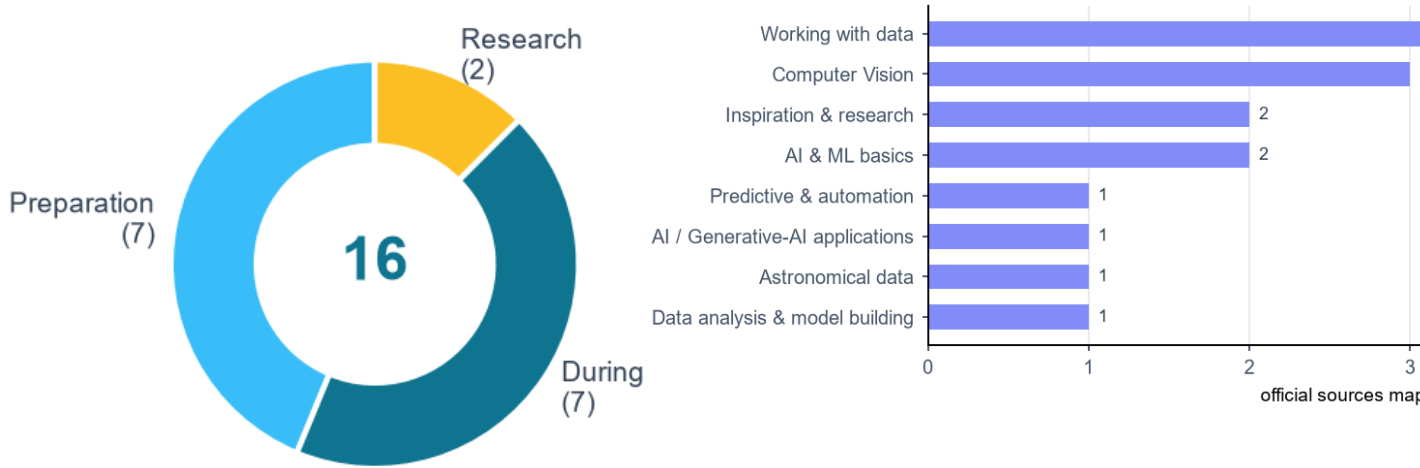


Figure 6 - The official source list at a glance. Left: when each source is meant to be used - before the event (preparation), during the build, or as research inspiration afterwards. Right: how the 16 sources spread over the eight required conditions; 'Working with data' is the most heavily covered, which matches a hackathon where the first job is always getting usable numbers.

15.1 The eight conditions

Each condition is listed with its Arabic wording from the brief, the source ids that back it, and the evidence inside this repository.

#	Condition	How the sources help (from the brief)	Ids	Evidence in this repository
1	AI & Machine-Learning basics ai_basics	فهم المفاهيم الأساسية واختيار الأسلوب المناسب للمشكلة. و Google ML Crash Course الفرق على تساعد المصادر العربية	S-01, S-02	The pipeline applies supervised-unsupervised fundamentals (scaling, contamination-based anomaly detection, train-free evaluation) and the council turns them into a scored verdict instead of a raw number.

#	Condition	How the sources help (from the brief)	Ids	Evidence in this repository
2	Working with data data_handling	عندما يحتاج الحل إلى بيانات فضاء أو رصد الأرض. البيانات ال أخرى أمثلة وبيانات حقيقية يمكن استخدامها Worldview NASA و Earthdata NASA توفر	S-03, S-04, S-05, S-10, S-11, S-12	Six live NASA/NOAA endpoints plus three reference CSVs; every row keeps its source endpoint and fetch timestamp in the traceability record.
3	Data analysis & model building analysis_modelling	البيانات، تحليلها، تدريب النماذج واختبارها بسرعة. أدوات مثل Google Colab و Scikit-learn لتجهيز يمكن استخدام	S-08, S-09	Cell-level completeness/accuracy audit, PCA (StandardScaler -> PCA) on real band matrices, and a quality score that is recomputed on every run rather than asserted.
4	Computer Vision computer_vision	أو Detection أو Classification. Segmentation لتطبيق Satellite Deep Learning و مراجع Datasets و Hugging Face استخدام Face إذا كان الحل يعتمد على الصور، يمكن استخدام	S-06, S-13, S-14	The shipped build ingests image *metadata* (FITS header cards, filters, exposure, WCS keywords) and raster night-light tiles, and labels them Level 1. Pixel-level segmentation/detection is the documented extension path, not a claim.
5	Astronomical data astronomical_data	استخدام Astropy وأدوات البيانات الفلكية المناسبة. إذا كان التحدي يحتاج إلى بيانات فلكية، يمكن	S-07	ICRS/GCRS coordinate handling, FITS header parsing, phase-folded light curves, synthetic spectra from NIST laboratory wavelengths, and a curated Hipparcos/Gaia/RC3 object catalogue.
6	AI / Generative-AI applications generative_ai	يعتمد على AI Generative أو LLMs عند الحاجة. الأدوات والنماذج المناسبة لبناء تطبيق أو نموذج أولي المصادر كمرجع لفهم البيانات والمشكلة، ثم اختيار يمكن لفريق استخدام	S-01, S-06, S-08	Three of four agents are LLM-backed with a deterministic fallback, every AI answer is tier-labelled, and the engine actually used is printed in the header pill.
7	Predictive & automation systems predictive_automation	تنبؤية وأتمتة أجزاء من الحل، حسب طبيعة المشكلة. نفس الموارد لفهم البيانات وبناء pipelines أو نماذج يمكن استخدام	S-03, S-05, S-09, S-12	One command runs ingest -> clean -> council -> citations -> UI; NOAA Kp is smoothed for trend reading, pass windows are derived from orbital period, and the health bar re-checks the whole frame each run.
8	Inspiration & research inspiration_research	ويمكن الاستفادة منهما للإلهام وليس كحلول جاهزة. أمثلة على كيفية تحويل مشكلة واقعية إلى حل تقني، توفر LightViz و FreeDSM	S-15, S-16	The light-pollution module follows the LightViz pattern (real imagery -> calibrated index -> human-readable scale) and is explicitly declared a *calibrated model*, not a photometric measurement.

Where each condition is applied

Condition	Files and modules that implement it	Status
1. AI & Machine-Learning basics	agents/agent1_ingestion.py - StandardScaler + IsolationForest agents/agent2_council.py - five-expert scoring, verdict logic static/mors.html - 5-tier AI protocol panel	core in build
2. Working with data	agents/agent1_ingestion.py - DONKI, NEOWS, Horizons, Kp agents/datasets.py - GIBS WMS VIIRS Black-Marble night lights agents/mors_data.py - APOD, Exoplanet Archive, CelesTrak	core in build
3. Data analysis & model building	agents/agent1_ingestion.py - cleaning log, quality audit agents/datasets.py - PCA reduction of the spectral matrix tools/build_report.py - report generation from live metrics	core in build

Condition	Files and modules that implement it	Status
4. Computer Vision	static/mors.html - ASTRONOMY.MORS image grid + FITS headers agents/mors_data.py images() - NASA APOD metadata + FITS cards agents/datasets.py - GIBS raster tiles for night lights	documented extension
5. Astronomical data	agents/agent1_ingestion.py - Astropy Time/SkyCoord transforms agents/mors_data.py - objects, FITS, photometry, light curves agents/mors_data.py - spectroscopy from NIST line lists	core in build
6. AI / Generative-AI applications	agents/agent2_council.py - Gemini Council of 5 agents/agent3_citations.py - Gemini citation enrichment agents/agent4_aimors.py - Ai.Mors conversational agent main.py - POST /api/mors/insight, POST /api/chat	core in build
7. Predictive & automation systems	main.py - four-agent pipeline with stage log and auto-run agents/agent1_ingestion.py - Kp 3-sample moving average agents/mors_data.py - satellite pass windows, quality monitor START.bat - environment check, port recovery, service launch	core in build
8. Inspiration & research	agents/datasets.py - night-light -> sky-brightness model PROJECTS.MORS - PRJ-002 Light-Pollution & Sky-Brightness Model problems P-002 / P-004 - calibration and tile-coverage findings	core in build

15.2 The official source list

Sixteen entries, exactly as supplied by the brief, each annotated with its phase, level, the Arabic guidance and the place in this codebase where it is used.

1. Google Machine Learning Crash Course [S-01]

Phase	Level	Kind	Condition
Preparation	Beginner -> Intermediate	Course	ai_basics

دورة Google المختصرة في تعلم الآلة

Goal فهم أساسيات تعلم الآلة بطريقة عملية.

You will learn Classification, Regression, Neural Networks, Data Preparation, Model Evaluation.

Who is it for الشخص الذي يعرف أساسيات Python ويريد تأسيس نفسه في ML.

When to use it كمرجع محدود أثناء الهاكاثون، وليس للتعلم من الصفر.

Link: <https://developers.google.com/machine-learning/crash-course>
Used in: agents/agent1_ingestion.py | agents/agent2_council.py

2. Machine Learning in Arabic (YouTube) [S-02]

Phase	Level	Kind	Condition
Preparation	Beginner	Video course	ai_basics

تعلم الآلة بالعربي — يوتيوب

Goal فهم تعلم الآلة باللغة العربية.

You will learn المفاهيم الأساسية قبل الانتقال إلى التطبيق.

Who is it for المبتدئين الذين يجدون المحتوى الإنجليزي صعبا.

When to use it لا تحتاج مشاهدة السلسلة كاملة إذا كان وقتك محدودا.

Link: no stable deep link supplied with the brief - open the platform directly.
Used in: static/mors.html - AI.MORS tier panel

3. NASA ARSET Training [S-03]

Phase	Level	Kind	Condition
Preparation	Beginner -> Advanced	Training	data_handling

تدريب NASA ARSET عن بعد

Goal

كيفية استخدام بيانات الاستشعار عن بعد وبيانات NASA في مشكلات حقيقية.

You will learn

Remote Sensing, Satellite Data, Earth Observation, تحليل البيانات.

Who is it for

الفرق التي ستحتاج بيانات رصد الأرض أو الأقمار الصناعية.

When to use it

قبل الهاكاثون لفهم المجال، والرجوع إليه أثناء الهاكاثون.

Link:

https://arset.gsfc.nasa.gov/

Used in:

agents/datasets.py | agents/mors_data.py

4. Inside Look at How NASA Measures Air Pollution [S-04]

Phase	Level	Kind	Condition
Preparation	Beginner	Case study	data_handling

نظرة داخلية: كيف يقيس NASA التلوث الهوائي (ARSET)

Goal

مثال عملي على استخدام بيانات الأقمار لدراسة مشكلة بيئية.

You will learn

كيف تستخدم بيانات NASA في مراقبة وتحليل التلوث.

Who is it for

من يريد فهم العلاقة بين بيانات الأقمار والمشكلات الواقعية.

When to use it

قبل الهاكاثون كمثال ميداني.

Link:

https://arset.gsfc.nasa.gov/

Used in:

agents/datasets.py - light-pollution pipeline

5. Machine Learning on Satellite Imagery (UN SDG Learn) [S-05]

Phase	Level	Kind	Condition
Preparation	Intermediate	Course	predictive_automation

تعلم الآلة على صور الأقمار الصناعية

Goal

الربط بين تعلم الآلة وصور الأقمار الصناعية.

You will learn

كيفية استخدام ML مع بيانات وصور الأقمار.

Who is it for

من لديه أساسيات ML ويريد تطبيقها على رصد الأرض/الفضاء.

When to use it

قبل الهاكاثون أو كمرجع أثناءه.

Link: no stable deep link supplied with the brief - open the platform directly.

Used in: agents/datasets.py - GIBS raster ingestion

6. Hugging Face Computer Vision Course [S-06]

Phase	Level	Kind	Condition
Preparation	Intermediate	Course	computer_vision

دورة Hugging Face في الرؤية الحاسوبية

Goal

تعلم استخدام الذكاء الاصطناعي في تحليل الصور.

You will learn

Image Classification, Object Detection, Segmentation, Deep Learning Models.

Who is it for

الفرق التي تفكر في حل يعتمد على الصور.

When to use it

قبل الهاكاثون أو كمرجع أثناءه.

Link: <https://huggingface.co/learn/computer-vision>

Used in: static/mors.html - image gallery | future: pixel models

7. Astropy Documentation [S-07]

Phase	Level	Kind	Condition
Preparation	Intermediate	Documentation	astronomical_data

توثيق Astropy

Goal

التعامل مع البيانات الفلكية باستخدام Python.

You will learn

FITS, coordinates, astronomical data.

Who is it for

الفرق التي تعمل على بيانات الفيزياء الفلكية وليس فقط رصد الأرض.

When to use it

قبل أو أثناء الهاكاثون حسب طبيعة المشروع.

Link: <https://docs.astropy.org/en/stable/>
Used in: agents/agent1_ingestion.py | agents/mors_data.py

8. Google Colab [S-08]

Phase	Level	Kind	Condition
During	All levels	Environment	generative_ai

بيئة البرمجة — Google Colab

Goal

كتابة وتشغيل Python و ML مباشرة من المتصفح.

You will learn

لا تحتاج إلى إعداد بيئة Python كاملة على الجهاز.

Who is it for

جميع الفرق التي تحتاج إلى برمجة.

When to use it

Analysis -> ML -> Visualization -> Prototyping.

Link: <https://colab.research.google.com/>
Used in: any notebook prototyping | main.py runs locally

9. Scikit-learn Documentation [S-09]

Phase	Level	Kind	Condition
During	Intermediate	Library	analysis_modelling

توثيق Scikit-learn

Goal

تنفيذ نماذج تعلم الآلة بسرعة.

You will learn

Classification, Regression, Clustering, Preprocessing, Evaluation.

Who is it for

أي فريق يستخدم ML على بيانات منظمة أو خصائص.

When to use it

ابحث عن الدالة أو الخوارزمية التي تحتاجها مباشرة.

Link: <https://scikit-learn.org/stable/>
Used in: agents/agent1_ingestion.py - StandardScaler, IsolationForest

10. NASA Earthdata Search [S-10]

Phase	Level	Kind	Condition
During	All levels	Data platform	data_handling

منصة اكتشاف البيانات — NASA Earthdata Search

Goal

البحث عن بيانات NASA المناسبة لمشروعك.

You will learn

استخدامه عندما تحتاج Dataset حقيقي بدل Dataset تجريبي.

Who is it for

الفرق التي تحتاج بيانات ببنية/أقمار/أرض.

When to use it

أثناء اختيار البيانات وأثناء تطوير المشروع.

Link: <https://search.earthdata.nasa.gov/>
Used in: agents/agent1_ingestion.py | agents/datasets.py

11. NASA Worldview [S-11]

Phase	Level	Kind	Condition
During	All levels	Visualisation	data_handling

NASA Worldview — تصور بيانات الأقمار

Goal

استكشاف صور وبيانات الأقمار الصناعية بصريا.

You will learn

اختيار المناطق، مقارنة التواريخ، فهم نوع الصور المتاح.

Who is it for

فرق رصد الأرض/الأقمار.

When to use it

أثناء البحث عن البيانات وتحديد فكرة المشروع.

Link: <https://worldview.earthdata.nasa.gov/>
Used in: agents/datasets.py - GIBS layer selection

12. NASA Nighttime Lights [S-12]

Phase	Level	Kind	Condition
During	Beginner	Dataset	data_handling

بيانات الإضاءة الليلية من NASA

Goal

الوصول إلى بيانات الإضاءة الليلية وتحليلها.

You will learn

مفيدة في Human Activity و Urbanization و Light Pollution.

Who is it for

الفرق التي تختار مشكلة مرتبطة بالملاحظات الليلية.

When to use it

أثناء الهاكاثون إذا كانت مناسبة لمشكلتك.

Link: <https://worldview.earthdata.nasa.gov/>
Used in: agents/datasets.py - VIIRS Black Marble tiles

13. Semantic Segmentation of Aerial Imagery (Kaggle) [S-13]

Phase	Level	Kind	Condition
During	Intermediate	Dataset	computer_vision

Kaggle — تجزئة دلالية لصور الأراضي

Goal

تجربة الرؤية الحاسوبية على صور جوية/فضائية.

You will learn

Computer Vision, Image Segmentation.

Who is it for

الفرق التي تحتاج Dataset سريع لتجربة النموذج الأولي.

When to use it

أثناء الهاكاثون.

Link: no stable deep link supplied with the brief - open the platform directly.

Used in: future: segmentation prototype

14. Satellite Image Deep Learning Techniques (GitHub) [S-14]

Phase	Level	Kind	Condition
During	Advanced	Technical reference	computer_vision

GitHub — تقنيات التعلم العميق لصور الفضاء

Goal

البحث عن تقنيات وأمثلة تعلم عميق لصور الفضاء.

You will learn

Computer Vision, Classification, Detection, Segmentation.

Who is it for

الفرق التي وصلت إلى مرحلة التنفيذ.

When to use it

استخدم البحث عندما تحتاج تقنية معينة، لا تقرأه كاملاً.

Link: no stable deep link supplied with the brief - open the platform directly.

Used in: future: detection / segmentation

15. FreeDSM (arXiv) [S-15]

Phase	Level	Kind	Condition
Research	Advanced	Research paper	inspiration_research

FreeDSM — ورقة بحثية على arXiv

Goal

مثال على مشروع يجمع بين إنترنت الأشياء والذكاء الاصطناعي.

You will learn

كيف تحول مشكلة واقعية إلى نظام يعتمد على AI.

Who is it for

الفرق المتقدمة التي تريد إلهاما أو فكرة عن البنية.

When to use it

قبل أو أثناء الهاكاثون. لا ينصح به كمصدر تعلم أساسي.

Link: no stable deep link supplied with the brief - open the platform directly.

Used in: PROJECTS.MORS - dossier architecture pattern

16. LightViz (arXiv) [S-16]

Phase	Level	Kind	Condition
Research	Advanced	Research paper	inspiration_research

LightViz — ورقة بحثية على arXiv

Goal

مثال على استخدام التكنولوجيا لمراقبة وتحليل التلوث الضوئي.

You will learn

يربط بين البيانات والمراقبة والتحليل في مشكلة واقعية.

Who is it for

الفرق التي تعمل على تلوث الضوء أو مشاريع مشابهة.

When to use it

قبل أو أثناء الهاكاثون.

Link: no stable deep link supplied with the brief - open the platform directly.

Used in: agents/datasets.py | PRJ-002 Light-Pollution model

15.3 Live data sources used by this build

Not part of the hackathon list - these are the endpoints this repository actually calls at runtime, with the acquisition path recorded for each one.

Id	Source	Kind	Via	What it gives	Used in
D-01	NASA DONKI (CME / FLR)	REST API	api	159 CME records + solar flares	agent1_ingestion.fetch_donki
D-02	NASA NEOWS	REST API	api	36 near-Earth objects, size / speed / miss distance	agent1_ingestion.fetch_neows
D-03	JPL Horizons	REST API	api	1442 heliocentric vectors (Astropy transforms)	agent1_ingestion.fetch_horizons
D-04	NASA GIBS WMS (VIIRS Black Marble)	WMS raster	api	night-light tiles -> sky-brightness model, 11 sites	datasets.light_pollution
D-05	NOAA SWPC planetary K-index	JSON feed	api	Kp 0-9 series + 3-sample smoothing + flare join	datasets.kp_index
D-06	NASA APOD	REST API	api	8 images with FITS-style header cards	mors_data.images()
D-07	NASA Exoplanet Archive (TAP)	TAP/SQL	api	60 confirmed planets + habitability proxy	mors_data.exoplanets()
D-08	CelesTrak GP (NORAD 2LE)	TLE feed	api	50 objects balanced LEO/MEO/GEO + pass windows	mors_data._sat_api()
D-09	NIST Atomic Spectra Database	Reference	local_model	laboratory line list for synthetic spectra	mors_data.spectroscopy()
D-10	Hipparcos / Gaia / RC3 / Sharpless / JPL	Catalogues	local_model	12 curated objects with ICRS coordinates	mors_data.objects()
D-11	Google Gemini API	LLM API	ai_synthesis	Council of 5, citations, Ai.Mors answers (429 -> local)	agent2_council / agent3_citations / agent4_aimors
D-12	Local reference CSVs (./data)	CSV	local_model	light-pollution, spectral and metric reference tables	agent1_ingestion.ingest_local_docs

[TRACE] ASI Hackathon source registry + project data sources | 1.0.0 | curated registry -> condition mapping -> project evidence | static registry, no network calls | agent:mors | 2026-09-25 06:28:19

acquisition order: api -> ai_synthesis -> local_model. A source is only allowed to fall to the next rung when the rung above it failed, and the rung used is recorded in the payload.

16. Honest limitations

- Gemini free-tier quota (HTTP 429) degrades Agents 2/3/4 to the deterministic local engines. The run still certifies, but it is not LLM cross-verified. This is surfaced as problem P-001 on the dashboard rather than hidden.
- IsolationForest is configured with contamination=0.1, so roughly 10% of cells are flagged as outliers by construction. The outlier count is a control figure, not a discovery.
- Multi-spectral profiles are laboratory reference spectra, not imagery. The widget is labelled accordingly.
- The Kp index window is much shorter than the flare window, so correlations are restricted to the overlap interval.
- NASA GIBS night-light tile coverage is incomplete; missing tiles are reported rather than interpolated.
- Satellite pass windows are period-derived estimates, not TLE propagations (estimate=true).
- Photometry, light curves and spectra are analytic reference models with injected photon noise. No telescope exposure is claimed.
- Quantum results are classical NumPy statevector simulations. No QPU was contacted.
- Citation graph depth is held at the deterministic baseline while the quota is exhausted.

Why these are listed

A platform that hides its own degradation cannot be audited. Each item above is also present as a machine-readable problem card in `/api/mors/problems`, with evidence, impact, root cause, analysis, solution, suggestions, an explicit confidence caveat and a metric-derived priority - so the dashboard, the API and this report all say the same thing.

Report generated 2026-09-25T06:28:20+00:00 from `http://127.0.0.1:5000` - run `run-20260925-061746-46a7` - verdict CERTIFIED 96/100.